BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 094 · Lesson 094 · CPU Performance: Clock, Cores and Cache

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a cache hit?

Q2. What is a cycle?

Q3. What is a core?

Q4. A search space of 64 candidates is repeatedly halved to one. How many halvings?

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I Do · Read new knowledge

Key vocabulary

GHz — Billions of cycles per second.

core — CPU execution core.

cache hit — Data found in CPU cache.

IPC — Instructions retired per cycle (workload dependent).

New learning

Observe — Clock speed counts clock cycles per second; one cycle is not always one complete instruction. Frequency alone cannot predict program performance.

Explain — Multiple CPU cores improve performance most for work divided into independently executable threads; a strictly sequential task may gain little.

Evaluate — Cache is small, fast storage close to CPU. A cache hit avoids some slower RAM accesses; cache size, locality and hit rate influence results.

Worked example

Compare 2 GHz and 3 GHz clock rates under equal work per clock and no bottlenecks.

1. Explain the mechanism — 3 GHz offers 50% more cycles each second than2 GHz; this does not universally imply 50% faster program runtime. — The comparison requires equal microarchitecture and workload assumptions.

We Do · Practise with guidance

A single-threaded algorithm runs on a 2-core or 8-core CPU of equal per-core performance. Explain likely benefit.

Explain the mechanism

It may not be substantially faster from extra cores alone; only one thread is the bottleneck.

Parallel benefits need tasks/threads that can run concurrently.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain effects of clock speed, number of cores and cache on performance. [6]

Question 2

CPU X 2GHz and Y 3GHz have identical IPC; compute ratio of cycles/sec and one reason runtime may differ from ratio. [6]

Question 3

Recommend improvements for school video rendering versus a sequential exam-marking script. Discuss CPU performance factors. [6]

Exit Ticket · Answer before checking

A 4GHz processor always completes any program twice as fast as a 2GHz processor.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Requested item found in cache.

Q2. A unit of processor clock timing.

Q3. An independent CPU execution unit.

Q4. 64→32→16→8→4→2→1; six halvings.

You Do pages · Independent questions

Q1. Clock affects cycles/sec [2], cores parallel tasks [2], cache reduces slower memory accesses [2].

Q2. 3/2=1.5 [2], Y50% more cycles/s [2], memory stalls/parallelisation or instruction mix [2].

Q3. Rendering often exploits cores [2], sequential script needs stronger single-thread performance [2], cache/IPC/workload caveats [2].

Exit Ticket · Model answer

False. Clock speed alone does not determine total performance.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: